Cheat sheet

Fine-tuning

SFT, PEFT, LoRA, alignment and evaluation

When

Fine-tune for	Style, format, narrow skill, smaller/faster model, consistent behaviour
Don't fine-tune for	Fresh or frequently changing facts (use RAG)

Methods

SFT	Train on prompt → ideal response pairs
Full fine-tuning	Update all weights; expensive
PEFT	Update few parameters (LoRA, prefix tuning, adapters)
LoRA	Low-rank matrices A·B added to weight updates; rank r, alpha, target modules
QLoRA	LoRA on 4-bit (NF4) quantised base model
Distillation	Student mimics teacher outputs
Preference	RLHF (PPO with reward model), DPO, ORPO, KTO

Data

Format	JSONL chat messages; consistent system prompts
Size	Hundreds to thousands of high-quality varied examples
Split	Hold out a test set; deduplicate; remove PII

Hyperparameters

Learning rate	~1e-4 to 2e-4 for LoRA (lower for full fine-tunes)

Epochs	1–3; watch validation loss
Rank r	8–64; higher = more capacity

Risks

Overfitting	Memorises examples; poor generalisation
Catastrophic forgetting	Loses general abilities
Safety drift	Fine-tuning can weaken refusals; re-run safety evals

Pro tip

Free practice: Google Colab or Kaggle notebooks (free GPU quotas) with Hugging Face TRL / PEFT or Unsloth on a 0.5B–3B open model.